



Ahsanullah University of Science & Technology

Department of Computer Science & Engineering

Course No : CSE 2214
Course Title : Assembly Language Programming Sessional
Assignment No : 05

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Submitted To : Ms. Sumaiya Nuha Mustafina & Ms. Tasnuva Binte Rahman

Submitted By-

Group : A₁
Name : Sakif Ahmed Rafi
Id : 00724105101024
Section : A

Question:

Write an assembly program to implement the following operations using the seven segment display and LEDs available in the emulation kit.

- Implement a three digit counter that will start from the last three digits of your student ID and ends at 700, increasing by 3 at each step. For example, if the last three digits of your ID is 054; the expected output:

054 -> 057 -> 060. ..696 -> 699

- Simultaneously, turn ON the LEDs two at a time in a reverse sequential order.

The sequence should be:

LED 7,8 -> LED 5,6 -> LED 3,4 -> LED 1,2 -> LED 7,8 ->.....

At any particular time, only two LEDs should be ON. After the first two LEDs are turned ON, the sequence should restart from LED 7 and 8.

Answer:

;00724105101024

L1:

```
MOV BX,VAL
MOV TEMP,BX
;2030H
XOR DX,DX
MOV AX,VAL
MOV BX,100
DIV BX
MOV VAL,DX
```

```
LEA SI,DIGITS
ADD SI,AX
MOV DX,2030H
MOV AL,[SI]
OUT DX,AL
```

```
;2031H
XOR DX,DX
MOV AX,VAL
MOV BX,10
DIV BX
MOV VAL,DX
```

```
LEA SI,DIGITS
ADD SI,AX
MOV DX,2031H
MOV AL,[SI]
OUT DX,AL
```

```
;2032H
MOV AX,VAL
```

```
LEA SI,DIGITS
ADD SI,AX
MOV DX,2032H
MOV AL,[SI]
OUT DX,AL
```

```
;LED
LEA SI,LED
ADD SI,LED1
MOV AL,[SI]
MOV DX,2070H
OUT DX,AL
```

```
CALL DELAY
```

```
DEC LED1
CMP LED1,0
JG SKIP
MOV BX,4
MOV LED1,BX
```

```
SKIP:
MOV AL,00000000B
OUT DX,AL
```

```
;LOOP CONTROL
MOV BX,TEMP
ADD BX,3
MOV VAL,BX
CMP VAL,700D
JLE L1
```

```
VAL DW 24
TEMP DW ?
LED1 DW 4
DIGITS DB 0111111B, 0000110B, 1011011B, 1001111B, 1100110B, 1101101B, 1111101B,
0000111B, 1111111B, 1101111B
LED DB 00000000B, 00000011B, 00001100B, 00110000B, 11000000B
```

```
DELAY PROC
MOV CX,50
L5:
LOOP L5
RET
DELAY ENDP
```